COMPUTER ENGINEERING

QUEZON CITY

| | | | |
|---|---|---|---|
| **COURSE CODE/TITLE:** | **CPE010** | **SECTION:** | **CPE21S4** |
| **DATE PERFORMED:** | 07/31/26 | **DATE SUBMITTED:** | |
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**ACTIVITY NO. ___ [ACTIVITY NAME]**

## 1. PROCEDURE OUTPUT

ILOAB

```
#include <iostream>
#include "stackList.h"

int main(){

    std::cout << "---- PUSHING 'c' 'p' 'e' ONTO THE STACK -----\n";

    // Push
    push('c');
    push('p');
    push('e');

    // Display The Current Top
    Top<char>(); //Top should be 'e'

    // Display All
    displayALL<char>();

    std::cout << "\n------ Popping all the elements ------\n";

    while(!isEmpty<char>()){
        std::cout << "Popped: "
                  << pop<char>()
                  << std::endl;
    }

    std::cout << "\n------ Testing the Underflow ------\n";

    pop<char>();

    return 0;
}
```

ILOAB2 > C stacklist.h > Top<T>()

```cpp
#ifndef STACK_LIST_H
#define STACK_LIST_H

#include <iostream>

// class node that contain data and the next
template <typename T>
class Node{
public:
    T data;     // VALUE THAT WE NEED STORE
    Node* next; // POINTER TO THE NEXT NODE
};

// global pointer for the top of the stack
template <typename T>
Node<T>* head = nullptr;

// operation for the stack

// PUSH
template <typename T>
void push(T newData){
    //create a newNode and assign the newData
    Node<T>* newNode = new Node<T>;
    newNode->data = newData;
    //new node points to current top, then becomes the new top
    newNode->next = head<T>;
    head<T> = newNode;
}

// POP
template <typename T>
T pop(){
    // check if the stack is empty:
    if(head<T> == nullptr){
        std::cout << "Stack Underflow." << std::endl;
        return T();
    }

    Node<T>* temp = head<T>;
    T tempVal = temp->data;

    head<T> = head<T>->next;

    delete temp;

    return tempVal;
}
```

```
49
50  // isEmpty
51  template <typename T>
52  bool isEmpty(){
53      return head<T> == nullptr;
54  }
55
56  // TOP
57  template <typename T>
58  void Top(){
59
60      if(isEmpty<T>()){
61          std::cout << "Stack Underflow." << std::endl;
62          return;
63      }
64
65      std::cout << "Top of Stack: "
66                << head<T>->data
67                << std::endl;
68  }
69
70  // Display All
71  template <typename T>
72  void displayALL(){
73
74      if(head<T> == nullptr){
75          std::cout << "Stack Underflow\n";
76          return;
77      }
78
79      std::cout << "Displaying the stack:\n";
80
81      Node<T>* temp = head<T>;
82
83      while(temp != nullptr){
84          std::cout << temp->data << std::endl;
85          temp = temp->next;
86      }
87
88      std::cout << std::endl;
89  }
90
91  #endif
92
```

OUTPUT

## 2. ANSWERS TO SUPPLEMENTAL ACTIVITY

**ILO C: Solve problems using an implementation of stack:**

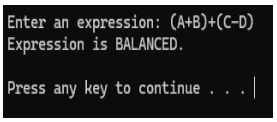
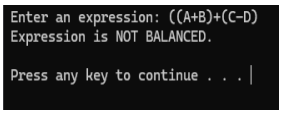
The following problem definition and algorithm is provided for checking the balancing of symbols. Create an implementation using stacks. Your output must include the following:

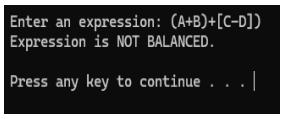
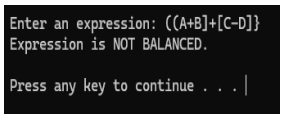
a. Stack using Arrays
b. Stack using Linked Lists
c. (Optional) Stack using C++ STL

**Problem Definition:**
Stacks can be used to check whether the given expression has balanced symbols. This algorithm is very useful in compilers. Each time the parser reads one character at a time. If the character is an opening delimiter such as (, {, or [- then it is written to the stack. When a closing delimiter is encountered like ), }, or ]-the stack is popped. The opening and closing delimiters are then compared. If they match, the parsing of the string continues. If they do not match, the parser indicates that there is an error on the line.

**Steps:**
1. Create a Stack.
2. While(end of input is not reached) {
a) If the character read is not a symbol to be balanced, ignore it.
b) If the character is an opening symbol, push it onto the stack.
c) If it is a closing symbol:
i) Report an error if the stack is empty.
ii) Otherwise, pop the stack.
d) If the symbol popped is not the corresponding opening symbol, report an error.
3. At the end of input, if the stack is not empty: report an error.

**Self-Checking:**
For the following cases, complete the table using the code you created.

| Expression | Valid? (Y/N) | Output (Console Screenshot) | Analysis |
|---|---|---|---|
| (A+B)+(C-D) | Y |  | This figure is balanced because both have an open and close parenthesis |
| ((A+B)+(C-D) | N |  | This figure is not balanced because one open parenthesis is unmatched |

| | | | |
|---|---|---|---|
| ((A+B)+[C-D]) | N |  | This figure is not balanced because one open parenthesis is unmatched |
| ((A+B]+[C-D]} | N |  | This figure is not balanced because of the bracket, open parenthesis, and curly bracket are unmatched |

**ILOC:**

```cpp
#include <iostream>
#include <string>
using namespace std;

#define MAX 100

char stack[MAX];
int top = -1;

bool isEmpty() {
    return top == -1;
}

bool isFull() {
    return top == MAX - 1;
}

void push(char ch) {
    if (!isFull())
        stack[++top] = ch;
}

char pop() {
    if (!isEmpty())
        return stack[top--];
    return '\0';
}

char peek() {
    if (!isEmpty())
        return stack[top];
    return '\0';
}

bool isMatching(char open, char close) {
    return (open == '(' && close == ')') ||
           (open == '[' && close == ']') ||
           (open == '{' && close == '}');
}

bool isBalanced(string exp) {
    top = -1;

    for (char ch : exp) {

        if (ch == '(' || ch == '[' || ch == '{') {
            push(ch);
        }
```

```
49
50          else if (ch == ')' || ch == ']' || ch == '}') {
51
52              if (isEmpty())
53                  return false;
54
55              char open = pop();
56
57              if (!isMatching(open, ch))
58                  return false;
59          }
60      }
61
62      return isEmpty();
63  }
64
65  int main() {
66
67      string expression;
68
69      cout << "Enter an expression: ";
70      getline(cin, expression);
71
72      if (isBalanced(expression))
73          cout << "Expression is BALANCED." << endl;
74      else
75          cout << "Expression is NOT BALANCED." << endl;
76
77      return 0;
78  }
```

```
Enter an expression: (A+B)+(C-D)
Expression is BALANCED.

Press any key to continue . . .
```

```
Enter an expression: ((A+B)+(C-D)
Expression is NOT BALANCED.

Press any key to continue . . .
```

```
Enter an expression: (A+B)+[C-D])
Expression is NOT BALANCED.

Press any key to continue . . .
```

```
Enter an expression: ((A+B]+[C-D]}
Expression is NOT BALANCED.

Press any key to continue . . .
```

**CONCLUSION:**

For this activity, I learned how to use the stack and how to check whether the symbols is an expression of balanced and learned also how to balance each symbols using the stack and array.

# NOT DONE

## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

**Tools Used:** _____

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

**Purpose of Use:** _____

*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

**Extent of Use:** _____

*Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.*

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I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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